



Bhavya Sree Ganja
Sr. Data Engineer

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PROFESSIONAL SUMMARY

- Senior Data Engineer with 7+ years of professional experience designing and building scalable, secure, and high-performance data platforms across Banking, Healthcare, and Retail industries.
- Strong background in PySpark, Spark SQL, Kafka, Snowflake, Databricks, DBT, and AWS, delivering cloud-native analytics solutions for batch, streaming, and metadata-driven workloads.
- Built a metadata-driven ingestion framework using AWS Glue, PySpark, DBT, and Amazon S3 that enabled rapid onboarding of 10+ enterprise source systems, reducing integration effort by 60%.
- Designed and deployed AWS-based ETL/ELT pipelines using S3, Glue, Lambda, Step Functions, Redshift, and CloudWatch for large-scale data processing, governance, and auditability.
- Proven expertise in Databricks, Delta Lake, Snowflake Streams/Tasks, and Lakehouse architecture to build scalable batch, CDC, and near real-time data pipeline
- Expert in SQL, Python, Spark, and DBT-based data modeling to build trusted analytical datasets and enterprise-ready reporting solutions.
- Led major legacy-to-cloud modernization initiatives, migrating SSIS, Hadoop, Hive, and MapReduce workloads to AWS and lakehouse platforms, improving ETL performance by 35–40%.
- Experienced in data modeling and warehouse architecture, including star schema, snowflake schema, fact/dimension modeling, SCD Type 2, and physical/logical data design.
- Developed and optimized Spark workloads using partition pruning, caching, broadcast joins, and AQE tuning, improving pipeline performance and cluster efficiency by 40%+.
- Skilled in relational databases including Oracle, PostgreSQL, MySQL, and SQL Server, along with NoSQL platforms such as MongoDB, DynamoDB, and HBase for large-scale data storage and access optimization.
- Hands-on experience building real-time streaming pipelines using Kafka, Spark Structured Streaming, AWS Kinesis, and Lambda to support fraud detection, AML analytics, and event-driven processing.
- Built secure multi-zone data lake architectures on Amazon S3 with AWS Glue Catalog, Lake Formation, KMS encryption, and VPC isolation to support HIPAA and PCI-compliant workloads.
- Developed enterprise analytics dashboards using Power BI, AWS QuickSight, Snowflake, and Redshift, enabling data-driven decision making across business teams.
- Integrated CI/CD pipelines using GitHub Actions, Jenkins, Bitbucket, CodePipeline, and Terraform, automating infrastructure provisioning and ETL deployment workflows.
- Engineered automated data quality and governance frameworks using Deequ, Great Expectations, and Python validation scripts, improving schema compliance and data reliability.

Education: Masters from Texas A&M University in Computer Science



TECHNICAL SKILLS

Programming Languages	Python, PySpark, Scala, Java, Bash, SQL
Tools	PyCharm, Visual Studio, Power BI, Tableau, Eclipse, Jupiter Notebook, Azure Databricks, Boto 3, Jenkins, Docker, Kubernetes.
Operating System	Windows, Unix, Linux
Databases	Oracle, MySQL, SQL, NoSQL (MongoDB), PostgreSQL, DynamoDB
Methodologies	Waterfall, Agile
ETL Tools	SSIS (SQL Server Integration Services), Apache Airflow, DBT
Cloud Services	AWS, Azure, Snowflake & Databricks
SCMs	Git, GitHub, Bitbucket, GitLab
Big Data Technologies	Apache Spark, Spark SQL, Kafka, Hadoop, Hive, HDFS, HBase, Sqoop
IAC Tools	AWS CDK, Terraform, CloudFormation

PROFESSIONAL EXPERIENCE

First National Bank, Pittsburgh, PA

Senior Data Engineer

Sep 2022 - Present

Responsibilities:

- Led modernization of legacy SSIS and Hadoop pipelines into a scalable AWS-based Databricks + Snowflake Lakehouse, enabling governed analytics for core banking domains including loans, deposits, payments, fraud, and regulatory reporting.
- Designed and built enterprise ETL/ELT pipelines using AWS Glue, PySpark, Spark SQL, Lambda, Step Functions, and DBT, automating ingestion from Finacle, Fiserv, and Salesforce source systems.
- Owned the complete DBT lifecycle including source configuration, staging models, marts, snapshots, schema tests, documentation, and CI/CD validation for enterprise analytics workloads.
- Engineered and optimized Snowflake data warehouse environments by configuring warehouses, resource monitors, RBAC, masking policies, and workload tuning, reducing compute cost by 25%.
- Designed CDC and SCD Type-2 frameworks using Snowflake Streams, Tasks, Delta Lake, and DBT, enabling near real-time reporting and historical data tracking for regulatory workloads
- Built a reusable Python-based ETL framework for validation, audit logging, retry handling, and automated alerting through SNS, PagerDuty, and CloudWatch, improving pipeline reliability and observability.
- Developed real-time streaming pipelines using Kafka and Spark Structured Streaming to process login events, ACH transfers, card transactions, and suspicious activity triggers for fraud and AML analytics.
- Migrated legacy Hive, HBase, Sqoop, and MapReduce workloads to AWS EMR and Glue, transforming batch-heavy pipelines into incremental near real-time processing workflows.
- Designed a secure multi-zone S3 data lake (Raw, Refined, Curated) integrated with Glue Catalog, Lake Formation, KMS encryption, and VPC endpoints to support governed enterprise data access.
- Built and orchestrated Airflow DAGs with dependency management, SLA monitoring, retries, and lineage tracking to ensure reliable pipeline execution and audit compliance.
- Delivered self-service data products and analytical marts with row-level security and masking for AML, Risk, Treasury, and Digital Banking teams, improving decision-making speed.



- Developed reusable Terraform modules to provision AWS infrastructure and Snowflake resources including databases, schemas, warehouses, roles, and grants, enabling fully automated deployments.
- Implemented robust CI/CD workflows using GitHub Actions and Jenkins, automating linting, unit testing, DBT validation, and Terraform plan checks for production releases.
- Established monitoring and observability using Datadog, CloudWatch, and Great Expectations, proactively detecting freshness issues, schema drift, SLA breaches, and data anomalies.
- Partnered with AML, Risk, Fraud, and Data Science teams to support fraud scoring pipelines, feature engineering workflows, and model deployment on Databricks.
- Supported early-stage Generative AI initiatives by preparing curated, high-quality datasets for LLM experimentation, enabling secure document search and contextual analytics for fraud and compliance teams.
- Collaborated with Data Science teams to build governed AI-ready data pipelines with masking, access control, and auditability, ensuring responsible use of enterprise data in GenAI workflows.
- Mentored junior engineers on PySpark optimization, Snowflake modeling, CI/CD, and AWS best practices, while supporting sprint planning and release governance.
- Optimized Oracle and PostgreSQL SQL workloads through indexing, partitioning, and incremental loading strategies, improving performance for finance and regulatory reporting.
- Integrated MongoDB-based customer interaction data into the lakehouse by implementing schema alignment, JSON flattening, and incremental synchronization for omnichannel analytics.
- Standardized modern data engineering practices using Delta Lake, DBT, Terraform, and Iceberg, reducing operational overhead and improving delivery consistency across banking workloads.
- Optimized Oracle and PostgreSQL SQL workloads through indexing, partitioning, and incremental loading strategies, improving performance for finance and regulatory reporting.

United Health Group, Irving, TX

Data Engineer

July 2020 – Aug 2022

Responsibilities:

- Designed and implemented CDC and SCD Type-2 pipelines using Snowflake Streams, Tasks, DBT, stored procedures, and Delta Lake, enabling longitudinal patient profiling, eligibility tracking, and near real-time risk analytics.
- Built scalable PySpark-based ETL frameworks for clinical and claims data harmonization, incorporating schema validation, PHI masking, encryption, and audit traceability for HIPAA-compliant processing.
- Developed real-time ADT event pipelines using AWS Kinesis, Lambda, and Spark Structured Streaming, triggering clinical alerts, care management notifications, and downstream outreach workflows.
- Created FHIR/HL7 standardization pipelines to transform raw EHR feeds into unified healthcare schemas, improving interoperability and regulatory reporting across provider networks.
- Built DBT analytical models supporting member segmentation, care gap analysis, chronic disease classification, medication adherence, and risk stratification for payer analytics.
- Automated data quality and governance checks using Deequ and Python validation frameworks, improving schema compliance, referential integrity, and member identity accuracy across healthcare datasets.
- Engineered governed Athena and Redshift analytical layers for HEDIS, Stars, claims cost, denial analytics, fraud/waste detection, and provider performance reporting.
- Designed secure multi-zone S3 data lake architecture (Raw, Staging, Curated, Analytics) integrated with Glue Catalog, Lake Formation, KMS encryption, and VPC controls for PHI-compliant storage and access.
- Integrated SageMaker scoring pipelines for high-risk patient identification, readmission prediction, and chronic care management, orchestrated using Glue, Lambda, and Step Functions.



- Optimized Spark and EMR workloads through executor tuning, partition pruning, shuffle optimization, and auto-scaling, reducing ETL runtime variance and compute cost by ~18%.
- Developed reusable CDC ingestion modules for claims, enrollment, pharmacy, and lab feeds with automated deduplication, timestamp reconciliation, and late-arriving data handling.
- Built and optimized Databricks Delta Lake pipelines for large-scale clinical, claims, lab, and pharmacy datasets, enabling incremental loads, schema evolution, and time-travel support for audit readiness.
- Developed modular Databricks notebooks and workflows using PySpark, Scala, DBT, and Delta Live Tables, supporting continuous eligibility updates, member 360 views, and near real-time enrichment
- Engineered secure Snowflake analytical environments with RBAC, Streams, Tasks, and zero-copy clones, enabling governed sandbox environments for actuarial and care analytics teams.
- Designed high-performance batch and streaming PySpark pipelines using partition pruning, broadcast joins, file compaction, and AQE tuning, improving large-scale clinical data processing across Databricks clusters.
- Built reusable Python and Scala utilities for ingestion validation, schema enforcement, audit logging, and exception handling, standardizing enterprise ETL governance patterns.
- Collaborated with clinical, informatics, actuarial, DevOps, and security teams, mentoring engineers on PySpark, AWS, DBT, and scalable Snowflake/Databricks best practices while supporting Agile delivery.

Walmart Global Tech, Virginia, VA

Data Engineer/ETL Developer

July 2018 – June 2020

Responsibilities:

- Developed enterprise-scale Hadoop-based data pipelines supporting customer analytics, operational reporting, and fraud-risk analysis across distributed data environments.
- Designed and implemented scalable ETL and ingestion frameworks using HDFS, Hive, Spark, Sqoop, and Shell scripting, enabling seamless integration of relational data sources including Oracle, PostgreSQL, and MySQL.
- Built PySpark and Scala-based batch pipelines for large-scale data transformation, cleansing, enrichment, and aggregation to support analytics and business intelligence workloads.
- Developed Spark and Scala ETL jobs to process high-volume clickstream and operational datasets, applying partitioning, checkpointing, and optimized transformations for batch and near real-time processing.
- Built automated data validation and ingestion workflows integrated with Hive Metastore and HDFS, improving schema consistency, metadata propagation, and reliable enterprise data onboarding.
- Designed Hive data models using partitioned, bucketed, and ACID-compliant tables, improving query performance and optimizing large-scale analytical workloads.
- Developed real-time streaming pipelines using Kafka and Spark Streaming to process clickstream and operational event data, supporting alerting and downstream analytics.
- Implemented Oozie-based workflow orchestration with dependency management, SLA monitoring, retry logic, and automated notifications to ensure reliable Big Data job execution.
- Managed HBase clusters for high-throughput, low-latency lookup workloads supporting feature engineering and operational analytics.
- Built Python and Bash automation utilities for metadata extraction, schema validation, file lifecycle management, and pipeline monitoring, reducing manual effort and improving data reliability.
- Orchestrated AWS EMR-based Spark and Hive workloads integrated with Amazon S3, implementing secure IAM, KMS encryption, and governed cloud storage for scalable data processing.
- Optimized Hive and Spark SQL workloads using partition pruning, caching, parallel execution, and efficient file formats such as Parquet and ORC, significantly improving query performance.
- Delivered pre-aggregated analytical datasets for Tableau and internal BI platforms, accelerating reporting cycles and enabling faster business decision-making.